

Upamanyu Ghose

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Education

DPhil in Psychiatry, University of Oxford (2023-present)

- Alzheimer's Research UK studentship.
- Applying machine learning to whole-genome sequencing and gene expression data to identify genetic variation linked to Alzheimer's disease.

MSc in Computer Science, University of Oxford (2019-2020)

- Oxford Weidenfeld-Hoffmann Scholarship.
- Awarded with Distinction.
- Key modules: Advanced Machine Learning, Computational Biology, Computational Game Theory.

BTech in Computer Science and Engineering, Manipal Institute of Technology (2015-2019)

- CGPA 9.86/10.0.
- Key modules: Machine Learning, Software Engineering, Data Structures and Algorithms.

Experience

Machine Learning Intern, Oxford Nanopore Technologies (2025)

- Developed a custom deep learning training criterion for DNA base-calling.
- Designed, implemented, and benchmarked it against the company's production criterion.

Research Assistant, Department of Psychiatry, University of Oxford (2020-2023)

- Developed GWANN, a neural network method to identify Alzheimer's-associated genes (first-author publication).
- Led the machine learning analysis on a type II diabetes drug discovery project at the Oxford Centre for AI in Precision Medicine (CAIPM), alongside structural chemistry and assay testing groups.
- Contributed to co-author publications on a proteomic ageing clock and efficient language models for biomedical text.
- Taught Python and machine learning at CAIPM summer schools.
- Built and maintained the lab's deep learning and bioinformatics servers.

Research Intern, Nanyang Technological University, Singapore (2018-2019)

- Developed PyTrack, an open-source Python toolkit for eye-tracking analysis (first-author publication).
- Led data collection for a study detecting emotion from eye-tracking sensors.
- Worked across computer science and psychology groups.

Volunteer, Teach Code for Good (2017-2018)

- Led curriculum planning and taught mathematics and programming to underprivileged school children in Manipal, India.

AI Team Member, Project Manas, Manipal Institute of Technology (2017-2018)

- Implemented a deep learning semantic segmentation algorithm for driverless vehicle navigation.
- Represented the team as a TEDx speaker.

Selected Publications

- **Ghose U**, Sproviero W, Winchester L, Amin N, Zhu T, Newby D, et al. Genome-wide association neural networks identify genes linked to family history of Alzheimer's disease. Briefings in Bioinformatics. 2025.
- Taylor N, **Ghose U**, Rohanian O, Nouriborji M, Kormilitzin A, Clifton DA, et al. Efficiency at scale: investigating the performance of diminutive language models in clinical tasks. Artificial Intelligence in Medicine. 2024.
- Argentieri MA, Xiao S, Bennett D, Winchester L, Nevado-Holgado AJ, **Ghose U**, et al. Proteomic aging clock predicts mortality and risk of common age-related diseases in diverse populations. Nature Medicine. 2024.
- **Ghose U**, Srinivasan AA, Boyce WP, Xu H, Chng ES. PyTrack: an end-to-end analysis toolkit for eye tracking. Behavior Research Methods. 2020.

Skills and Interests

- **Programming:** Python, R, Bash.
- **Machine learning:** Scikit-learn, LightGBM, SHAP, PyTorch.
- **Methods:** Supervised modelling, unsupervised modelling and clustering, neural networks, interpretable machine learning.
- **Languages:** English, Bengali, Hindi.
- **Sport:** Captain (2024-25), rower and coxswain, Green Templeton Boat Club.
- **Music:** Singer, Green Templeton College Choir.